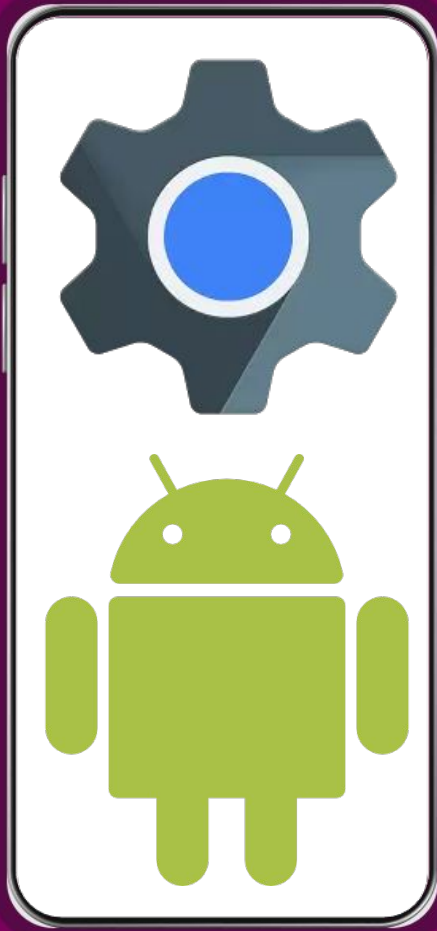
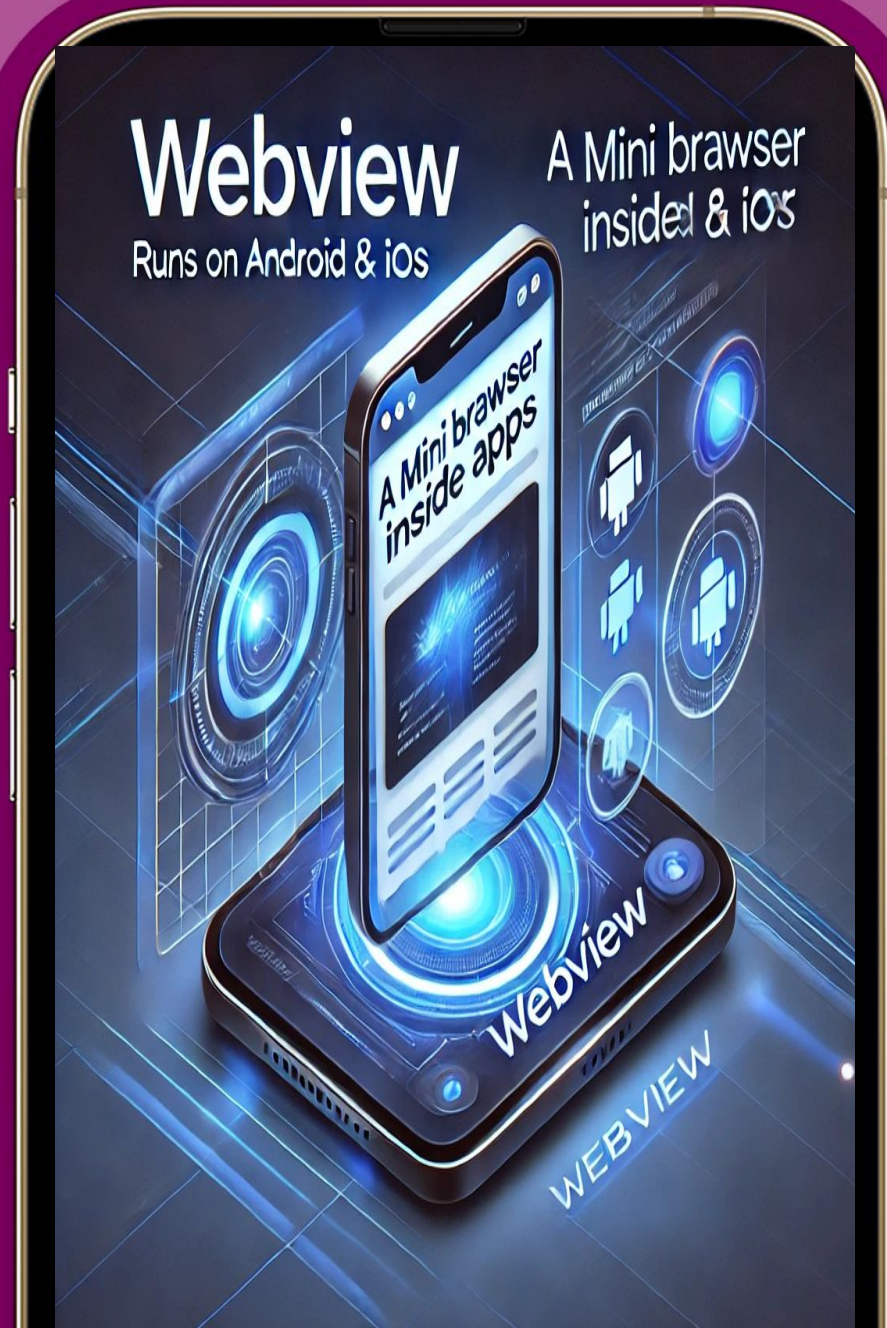




WEBVIEWS

- Understanding
- Advantages
- Use cases in Meesho



What is a Webview

- Webview is a mini browser which runs as a separate service in android or IOS Operating systems, which can be leveraged to mount websites inside of any android app.
- A webview can be used to display web content in any portion inside of an android app need not be a full screen display

Advantages

- Quick Experimentation - No need to go for app release cycle once the webview is integrated.
- Faster development and Deployments.
- Reduced App size.
- Same code - powers up android ,ios and web screens - reducing development effort by several folds.

Communication with Native App

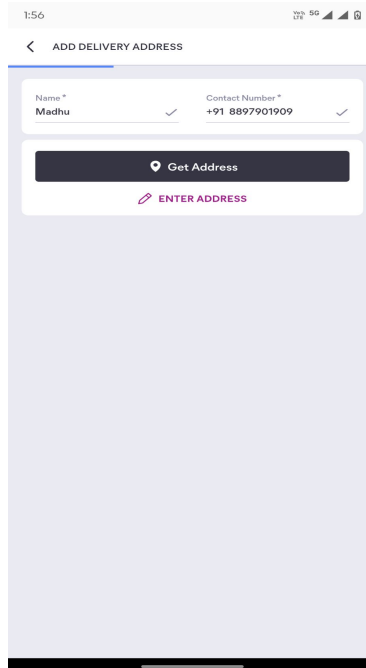
- There is a middleman (Webview Bridge) who is a broker between android and web runtimes.
The communications from web -> android or android -> web happens via this Webview Bridge
- Both Android and web components register event listeners to handle incoming messages.
- Example: User Config Bridge returns the config data saved in android to avoid redundant API calls.
`window.cartReview.sendConfigResponse()`

Webviews over SDUI

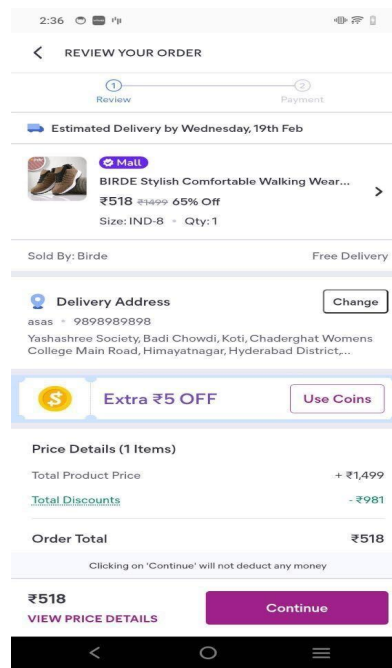
- SDUI still requires native updates when modifying UI logic beyond what is defined on the server. Hence If the native app does not support a new component in SDUI, a client update is required
- SDUI is limited to predefined components, making complex interactions harder to implement also limiting external library usage

Where all are we using webviews in Meesho already

Address Form



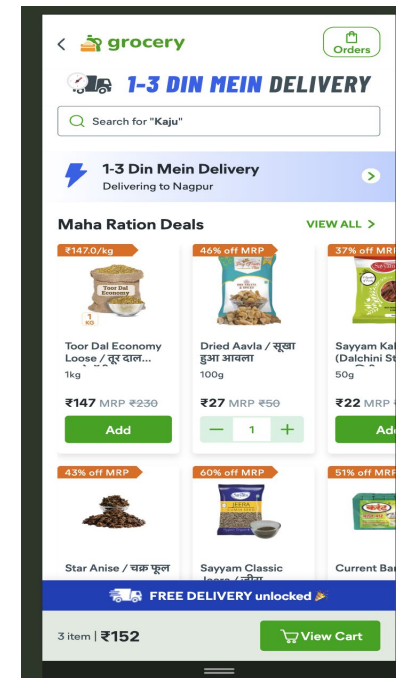
Cart/Review



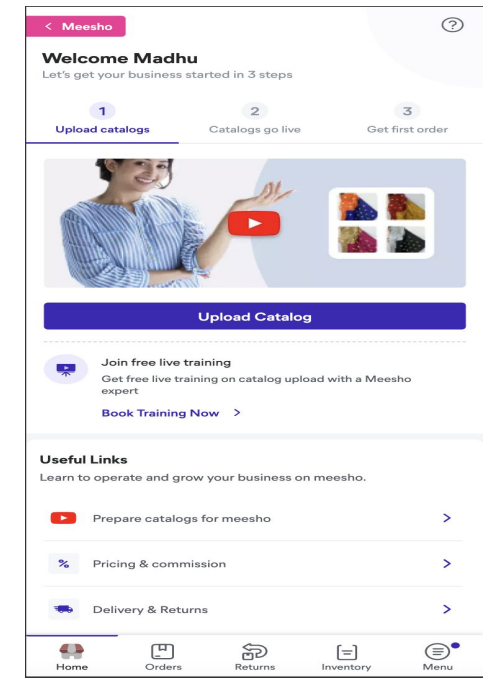
Reels



Meesho Grocery



Seller Panel



Disadvantages of Webviews

- Performance Issues: Higher loading time as content to load the webview is fetched from the internet.
- WebView issues on lower browser versions due to our user base primarily using low-end devices with outdated browsers.

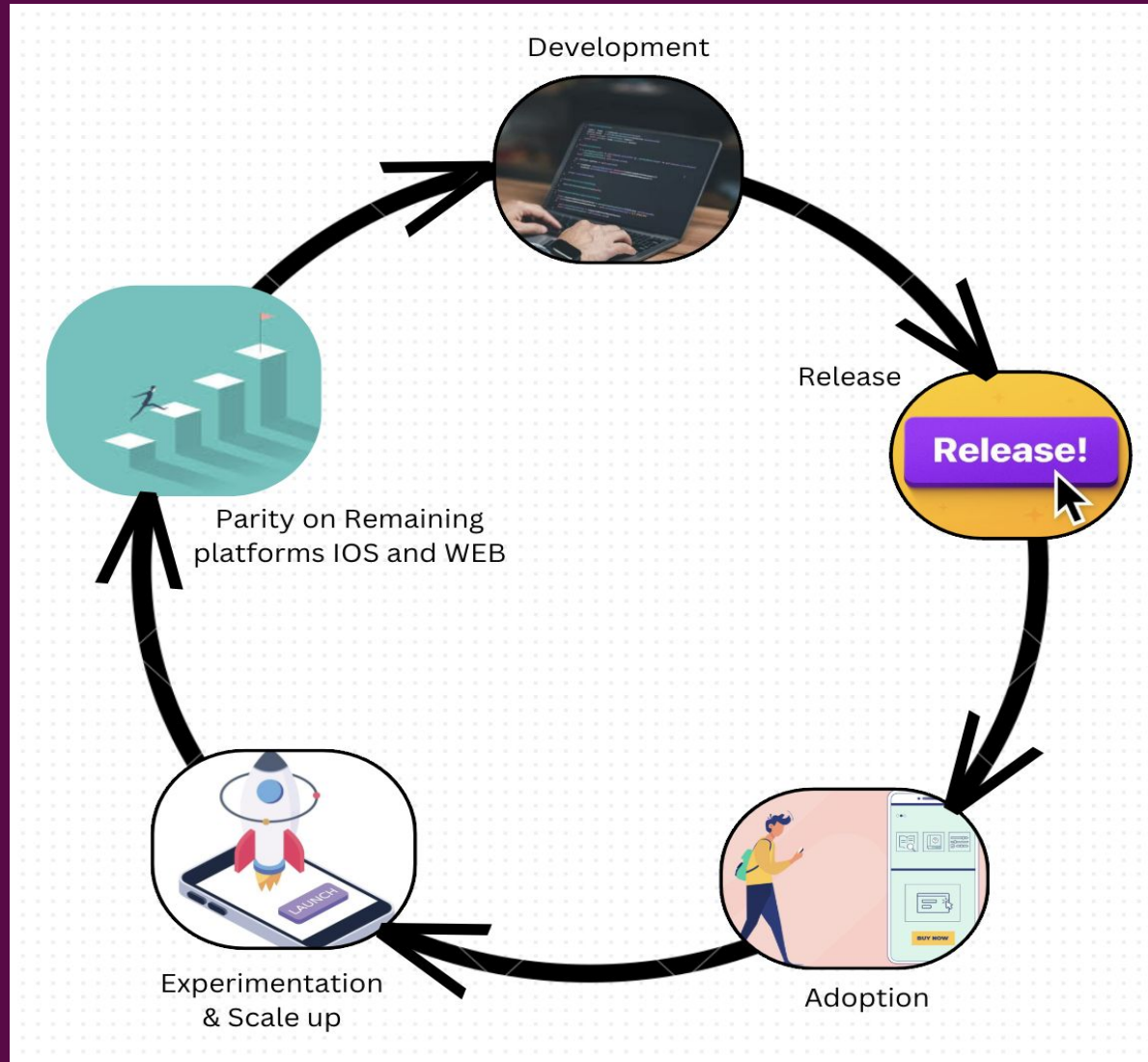
Some modern JS features may not work on outdated WebView versions.

Optimisations for webviews

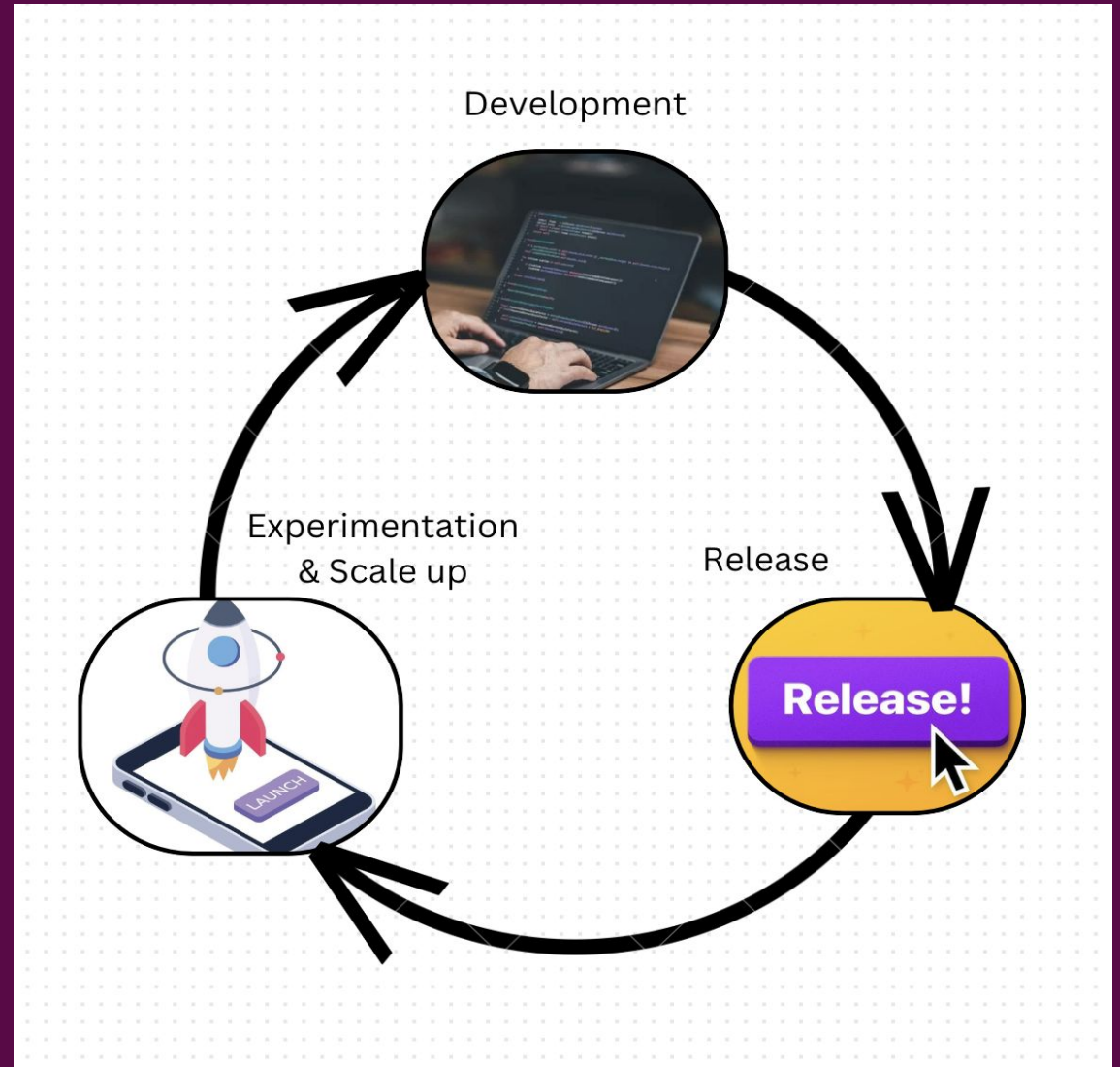
- Since the webview runtime is with android, We made android have all the pre-downloaded cache of the web assets (Html, JS, css, images etc) beforehand and it intercepts all the network requests going via webview instance and respond back with cached asset.
Using this we could achieve the native feel for android with very/less to no loading time.
- The monorepo setup with lerna helped us spin up as many webviews as we want.
This setup offers below advantages
 - Code Reusability
 - Multi build setup for webviews
- Reduce JS footprint as much as possible
 - Use only needed 3p libraries
 - Find and use lightweight alternatives for the 3p libraries which have large JS bundle sizes

App v/s Webview release Cycle

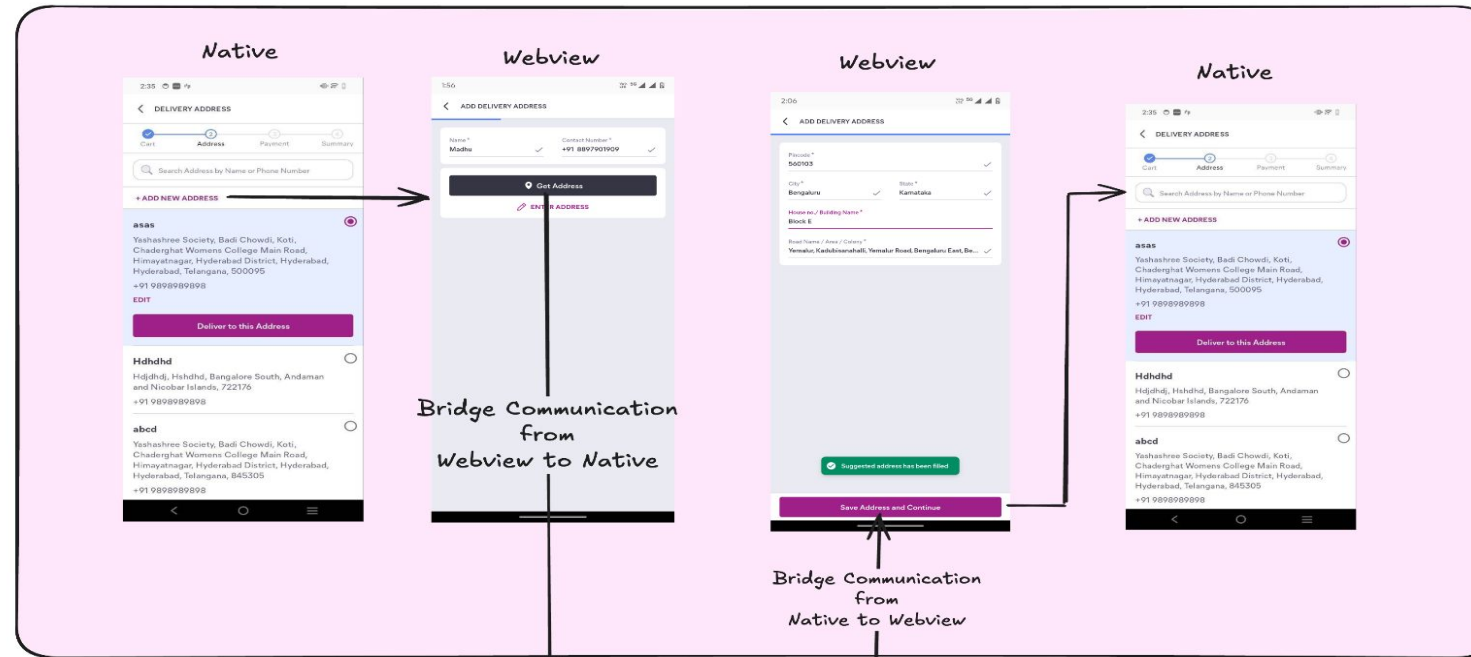
Development Cycle with Native



Development Cycle with Webview

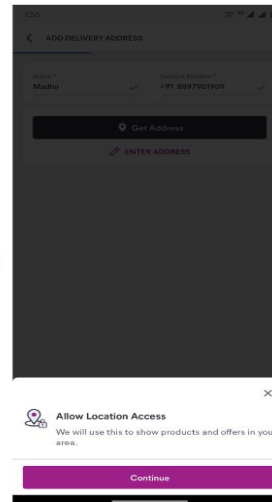


Webviews integration with Android



Here `getMyLocation` is an Android exposed function which web calls to get the location data

```
window.addEditAddress.getMyLocation()
```



Here `onFetchMyLocation` is a Web exposed function which android calls with location details as params

```
(window.addEditAddress?.onFetchMyLocation({ lat: 17.385, long: 78.4867, accuracy: 19 })))
```

